

5 G Code and M Code

G code	Description	Description and Example
G0	Rapid positioning	G0 X..Y.. Z.. , move as quickly as possible to a given point. In our system the G0 speed is defined by #103.
G01	Linear interpolation	G1 X... Y... F... , move to a given point by the F speed.
G02	Clockwise circular interpolation	By Radius: G2X..Y..Z..R..F.. By the center of a circle: G2X..Y..Z..I..J..K..F
G03	Counterclockwise circular interpolation	By Radius: G3X..Y..Z..R..F.. By the center of a circle: G3X..Y..Z..I..J..K..F
G04	Pause for a given duration of time	G4 P10000 pause for 10000 millisecond
G15	Switch back to Cartesian coordinates.	G16X10Y90: Moves to (0,10) Y180: Moves to (-10,0) Y270: Moves to (0,-10) Y0 : Moves to (10,0)
G16	Enable Polar Coordinates programming mode X represents the Distance and Y represents the Angle	G15X0Y0: Moves to (0,0) and Polar Coordinates OFF.
G17	XY Plane select	G17, all commands are now to be interpreted in the XY plane
G18	XZ plane select	G18, all commands are now to be interpreted in the XZ plane
G19	YZ plane select	G19, all commands are now to be interpreted in the YZ plane
G20	Inch mode	passed coordinates will be considered as Inches, so internally translated to millimeters
G21	Millimeter mode	passed coordinates will be considered as millimeters
G28	Go to the recorded origin position via a reference point at rapid speed	G91 G28 X10 Y0 Z0 The X axis will first move 10mm to the right, then the XYZ axes will go to the machine axis 0.
G30	Go to the recorded origin position via an alternative reference point at rapid speed	G90G30X_Y_Z_A_P_ P program cites 1,2,3, represents which reference point it will go
G31	Report current Z probe status	G31X_Y_Z_A_L_Q_K_F_ XYZA is the XYZA Probe trip L is the probe signal effective level K represents that disable or enable the hard limit during probing Q represents that when probe signal is triggered the stop mode: 0-Stop by Decelerate 1- Stop immediately F represents the Probing speed G91G31Z-1000L0Q1K0F100 Z axis probe down 1000mm and the signal effective leve is 0, disable the hard-switch and when probe signal is triggered the stop immediately.
G53	Move in absolute machine coordinate system	Need to Home each axis in advance, or system alarms and exits. G53X10Y10 XY Move to the mach coordiante position (10,10).
G54	G54 coordinate	G54, choose the active coordinate system G54
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G56	G56 coordinate	G56, choose the active coordinate system G56
G57	G57 coordinate	G57, choose the active coordinate system G57
G58	G58 coordinate	G58, choose the active coordinate system G58
G59	G59 coordinate	G59, choose the active coordinate system G59
G68	ENABLE Rotate program coordinate system	G68X10Y5R45 Rotate 45 degrees around (10,5)
G69	DISABLE Rotate program coordinate system	

G code	Description	Format and description
G73	Canned cycle - peck drilling	Need to configure the spindle as servo spindle. G73 X_Y_Z_R_Q_F_K_; Canned cycle - peck drilling, work with F command. X_Y_ : Hole Position, Z_ : The distance between R position and hole bottom. R_ : The distance between the initial position and R point; Q_ : The drilling depth of each feeding, P_ : Pause time, K_ : Repeated times
G74	Canned cycle - left hand rigid tapping	Need to configure the spindle as servo spindle. G74 CCW rotate to the hole bottom by F speed(Z), and CW rotate to R point. G84 CW rotate to the hole bottom by F speed(Z), and CWW rotate to R point. X_Y_ : Hole Position, Z_ : The distance between R position and hole bottom. R_ : The distance between the initial position and R point; P_ : Pause time G98/G99 G74 X_Y_Z_R_P_F_; G98/G99 G84 X_Y_Z_R_P_F_;
G84	Canned cycle - right hand rigid tapping	Example: M29S200 (Enter the rigid tapping mode. In the rigid tapping mode, the Z axis and servo spindle adopt the interpolation method movements.) G74X-10Y-20Z-30R5F500 (Thread pitch of tap is $F/S=500/200=2.5\text{mm}$) G84X10Y20Z-30R5F500 M5
G90	Absolute distance mode	In absolute distance mode, axis numbers (X, Y, Z, A) usually represent positions in terms of the currently active coordinate system.
G91	Incremental distance mode	In incremental distance mode, IJK numbers usually represent increments Incremental distance mode I from the current controlled point.
G92	Adjust current coordinate	G92X_Y_Z_A_ By adjusting the origin point of the current coordinate system, To make the current point have the coordinates as XYZ assigned.
G98	Initial level return after canned cycles	Retract perpendicular to the selected plane to the position that axis was in just before the canned cycle started
G99	R-point level return after canned cycles	retract perpendicular to the selected plane to the position indicated by the R word.
G128	Home command	G128X_Y_Z_A_ When programming word (X/Y/Z/A) set to 1, indicate that the according axis execute Home action. G128X1Y1Z1 XYZ axis execute Home action at the same time.

M code	Description	Description and Example
M0	Program Pause	To stop a running program temporarily
M3	Rotate spindle clockwise	To start the spindle turning clockwise at the currently programmed speed
M5	Stop spindle rotation	To stop the spindle from turning
M6	Tool change	M6 Start when the command is encountered. It will then wait for Cycle Start to be pressed
M8	Flood coolant on	Coolant output signal is valid.
M9	All coolant off	Coolant output signal is invalid.
M10	Lubrication ON	Lubrication output signal is valid.
M11	Lubrication OFF	Lubrication output signal is invalid.
M29	Spindle command group command	Enter rigid tapping mode , Call before G74/G84
M98	Call subroutine	To call a subroutine program within the current part program file code.
M99	Return from subroutine	To return from a subroutine program
M30	Program end	End all the program
M105	X axis Home command	If the commands appear on same line, then all the axis execute Home action at the same time. M107 : Z axis Home, M105M106: X axis and Y axis Home.
M106	Y axis Home command	
M107	Z axis Home command	
M108	A axis Home command	
M47	Repeat program from first line	After the program is finished, it will be executed from the beginning, and it will continue to cycle.